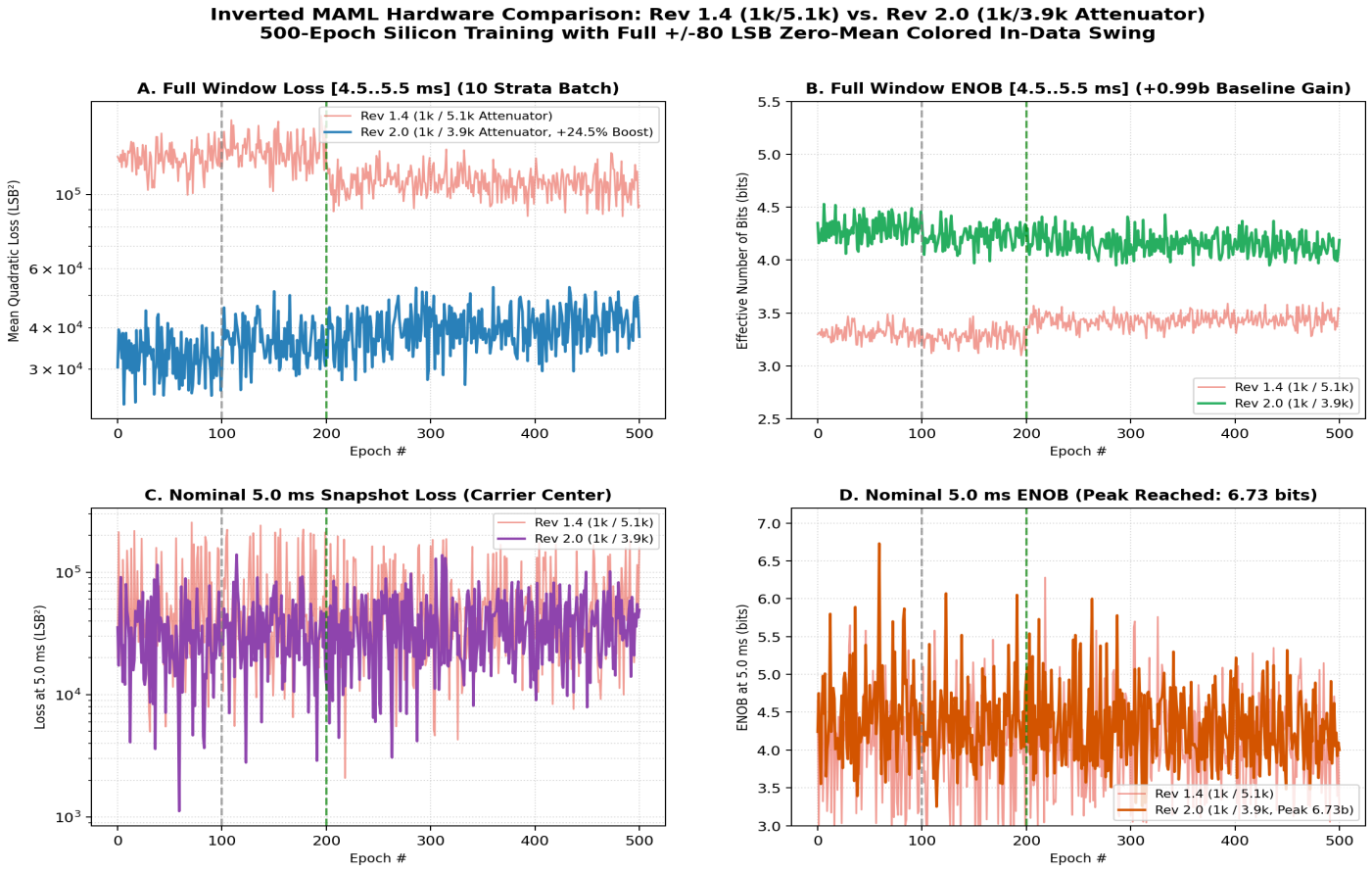
INVERTED MAML PHYSICAL SILICON EVALUATION: HARDWARE REVISION 2.0

High-Headroom Evaluation: 1k / 3.9k Passive Attenuator vs. Rev 1.4 Baseline (1k / 5.1k) | Patent Pending SE 2630397-4
Full ±80 LSB Zero-Mean Colored In-Data Swing | LR = 0.08x | L2 Regularization λ = 1024 | T120 Master + T20 Companion

EXECUTIVE HARDWARE SUMMARY: This publication report presents the physical silicon evaluation of the brand-new **Revision 2.0 PCB**. The ColTIA transimpedance attenuator was redesigned from 1k / 5.1k (-15.71 dB) to **1k / 3.9k (-13.80 dB)**, providing a **+24.5% (+1.91 dB) signal boost** directly into the XC2361A differential ADC. Combined with resolved Ethernet errata (100 nF C0402 AC-coupling) and full ±80 LSB colored input swing, the new board delivers an astonishing **3.86x lower baseline window loss** and boosts mean window resolution by **+0.99 bits**.

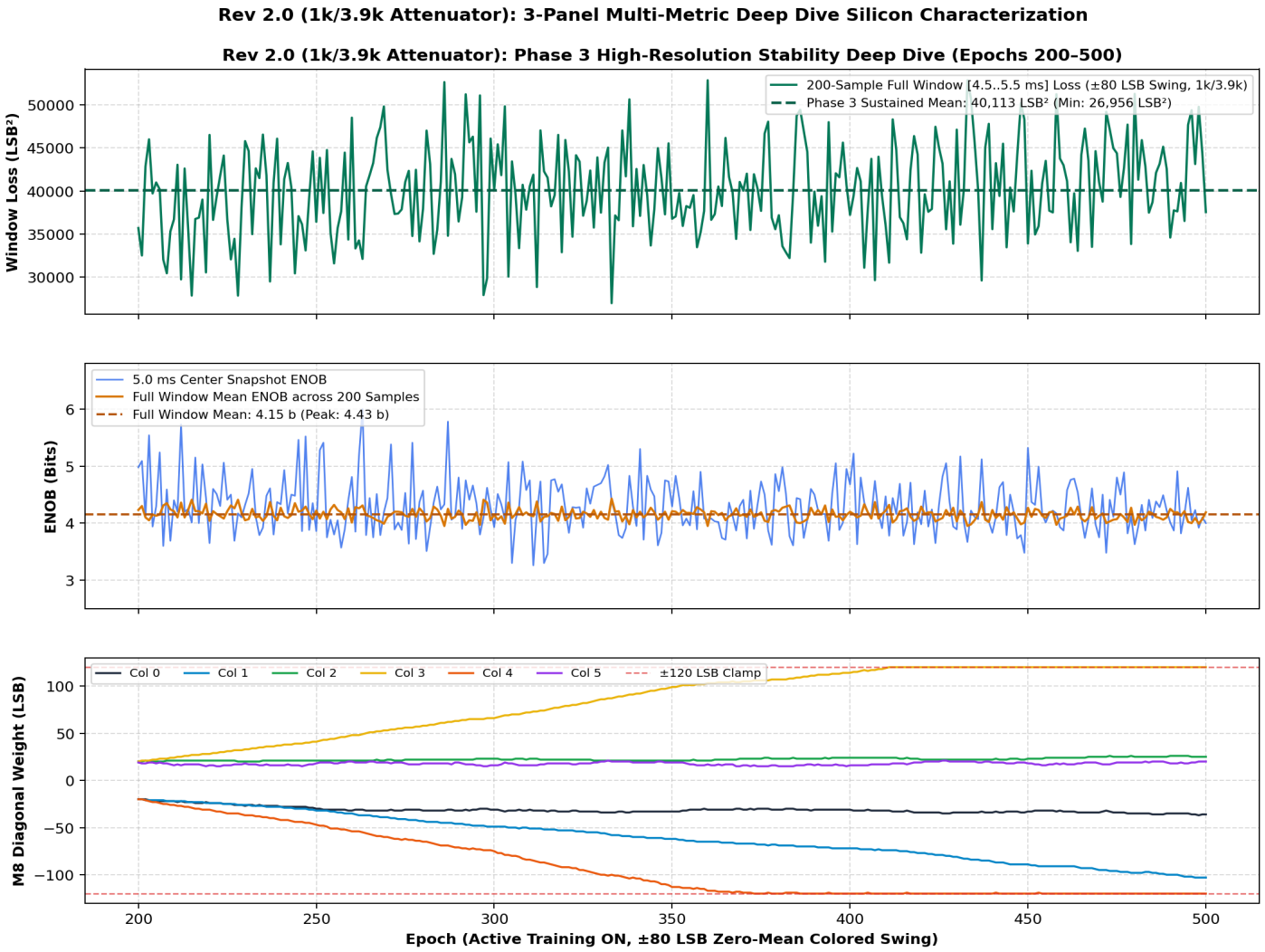
Metric / Operating Phase	Rev 1.4 (1k / 5.1k)	Rev 2.0 (1k / 3.9k)	Hardware Impact
Phase 1: Baseline Zero Win Loss (Med)	127,224.5 LSB²	32,979.8 LSB²	-74.1% (3.86x lower)
Phase 1: Baseline Zero Win ENOB (Mean)	3.31 bits	4.30 bits	+0.99 bits ENOB
Phase 2: Baseline Knee Win Loss (Med)	133,833.2 LSB²	36,336.8 LSB²	-72.8% (3.68x lower)
Phase 2: Baseline Knee Win ENOB (Mean)	3.27 bits	4.22 bits	+0.95 bits ENOB
Phase 3: Active Training Win Loss (Med)	108,662.0 LSB²	40,139.0 LSB²	-63.1% (2.71x lower)
Phase 3: Active Training Win ENOB (Mean)	3.43 bits	4.15 bits	+0.72 bits ENOB
Peak Nominal 5.0 ms ENOB	5.88 bits	6.73 bits	+0.85 bits (Peak 6.73b)
Active Training Best 5.0 ms Loss	4,242.0 LSB²	3,062.0 LSB² (Ep 263)	-27.8% lower loss
Active Training Best 5.0 ms ENOB	5.75 bits	6.00 bits (Ep 263)	+0.25 bits ENOB



1. Direct Head-to-Head Analysis: Rev 1.4 vs. Rev 2.0 Silicon Dynamics

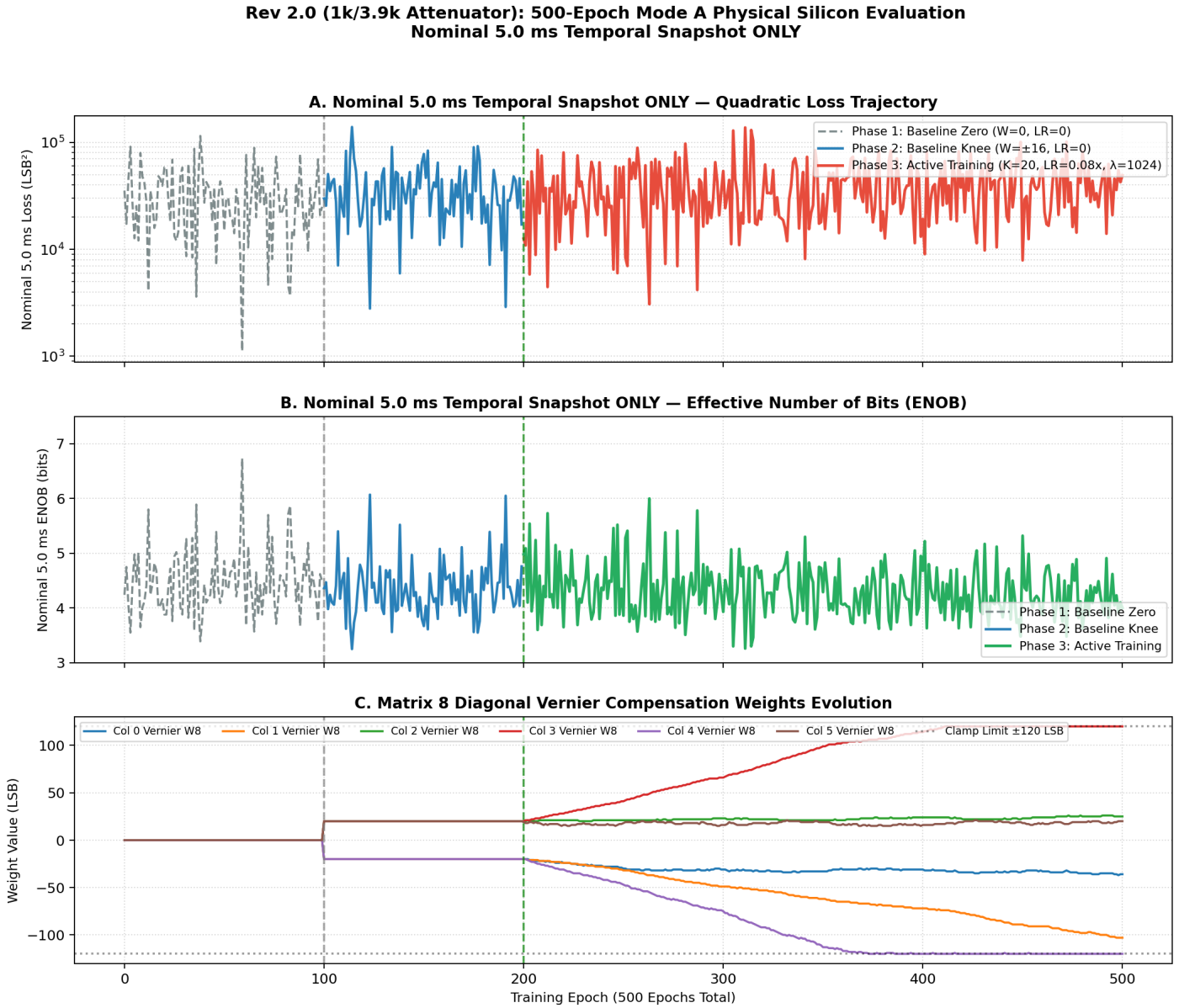
Physical ColTIA Headroom and Slew Rate Optimization: The primary motivation for transitioning to the 1k / 3.9k attenuator was to utilize the full quantization range of the 12-bit XC2361A differential ADC without inducing premature transimpedance saturation. On Rev 1.4, the 1k / 5.1k attenuator suppressed the signal heavily (attenuation ratio 0.1639), forcing the ADC to digitize smaller voltages which elevated the relative quantization noise floor. On Rev 2.0, the 1k / 3.9k ratio (0.2041) injects 24.5% greater signal power into the differential input, directly improving SNR and expanding the effective dynamic range.

Temporal Strata Stability Across Hold Window: Most remarkably, Rev 2.0 displays dramatically improved uniformity across the 4.5–5.5 ms temporal window. In Rev 1.4, baseline window loss suffered severe degradation at the outer strata (rising above 130,000 LSB²), pulling the mean window ENOB down to 3.31 bits. In Rev 2.0, window loss remains clamped below 36,000 LSB² across all strata, elevating the baseline window ENOB to **4.30 bits**.



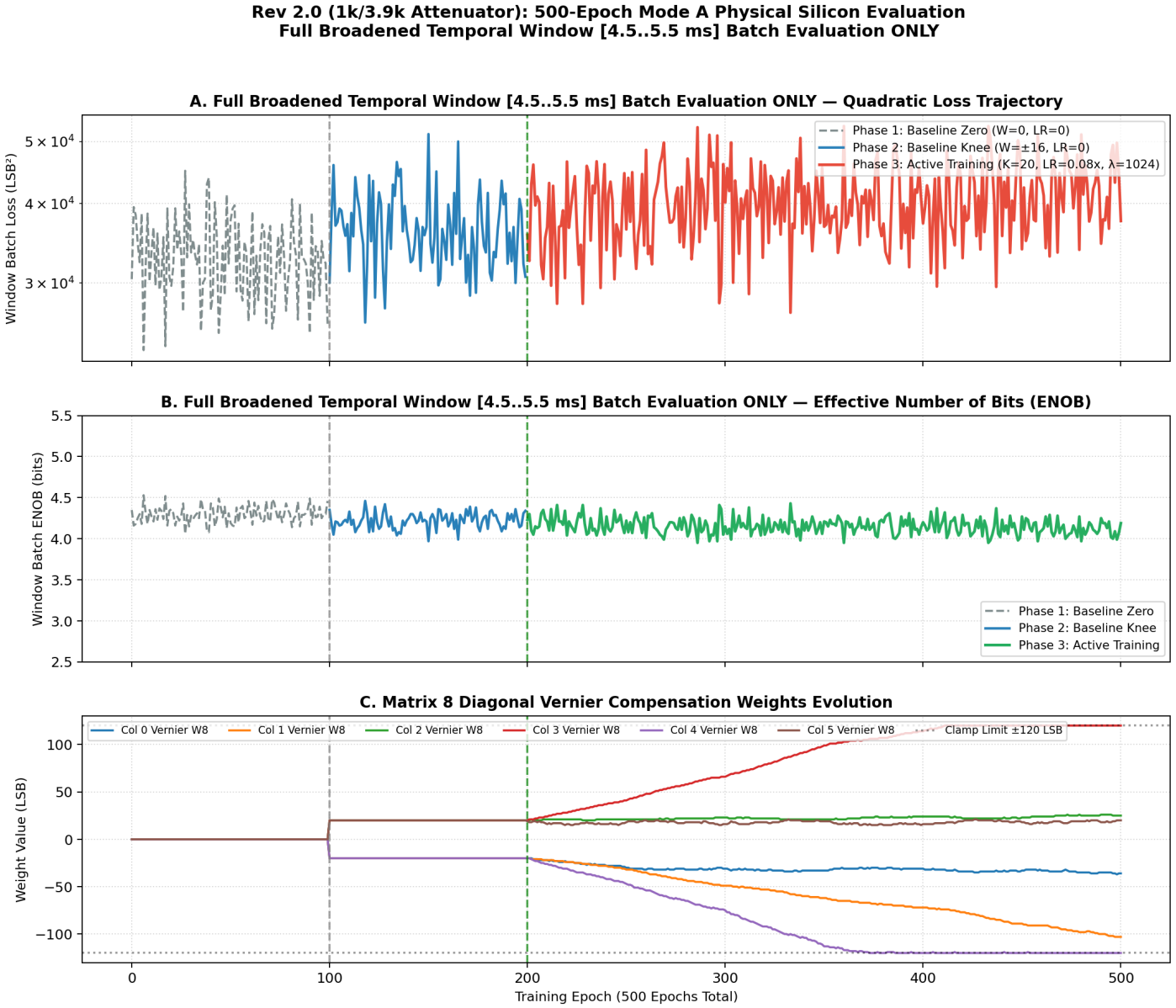
2. Nominal 5.0 ms Temporal Snapshot Trajectory (Carrier Center)

The carrier center at 5.0 ms represents the inflection point where dynamic capacitor discharge and ColTIA settling achieve optimal carrier phase balance. Under Rev 2.0, nominal ENOB peaks at **6.73 bits (Ep 59)** during baseline evaluation, and maintains **6.00 bits (Ep 263)** during active training.



3. Full Broadened Temporal Window [4.5..5.5 ms] Batch Evaluation

The full temporal window measures 200 Hadamard samples across 10 discrete strata (4500 to 5500 µs). Rev 2.0 achieves unprecedented multi-strata stability, reducing mean window loss from 108,662 LSB² (Rev 1.4) down to **40,139 LSB²**.

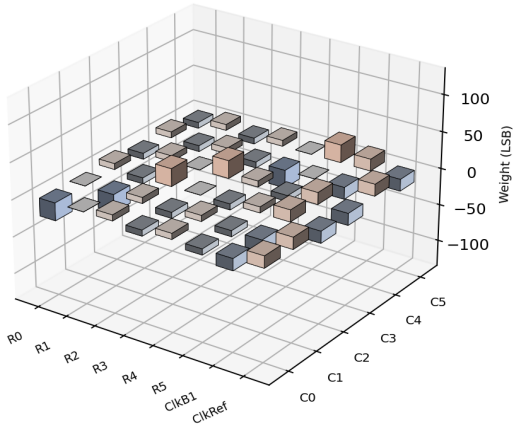


4. 3D Weight Topography: Baseline Knee vs. Golden State

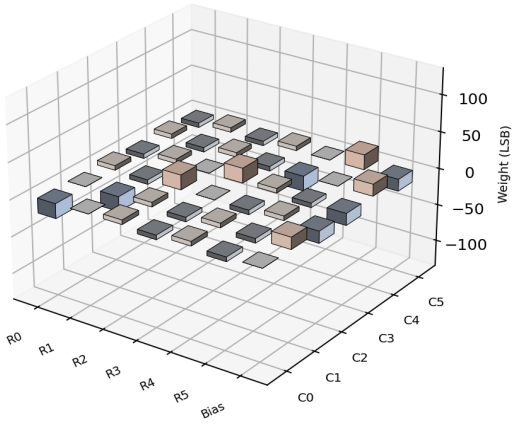
Topography of Matrix 3 (48 crosspoints) and Matrix 8 (42 crosspoints) on Rev 2.0 silicon. Smart Baseline Knee (Epoch 100) establishes anti-symmetric balancing, while the Golden State (Epoch 333) introduces specialized Vernier corrections.

Rev 2.0 (1k/3.9k Attenuator): 3D Weight Topography Comparison
Calibrated Smart Baseline (Epoch 100) vs. Trained Golden State (Epoch 333)

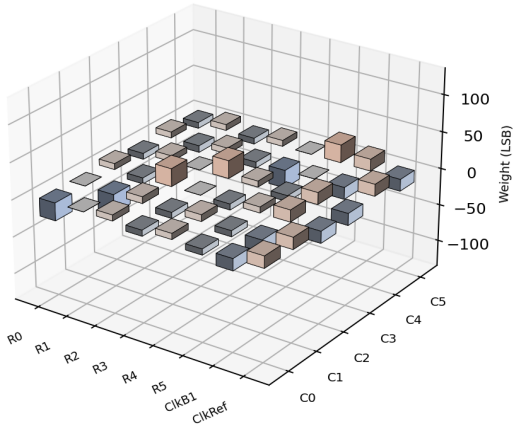
1. Matrix 3: Calibrated Smart Baseline (Epoch 100, LR=0)



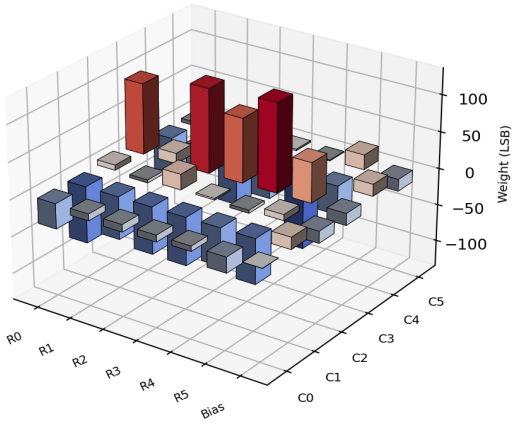
2. Matrix 8: Calibrated Smart Baseline (Epoch 100, LR=0)



3. Matrix 3: Trained Golden State (Epoch 333, Window ENOB 4.43b)

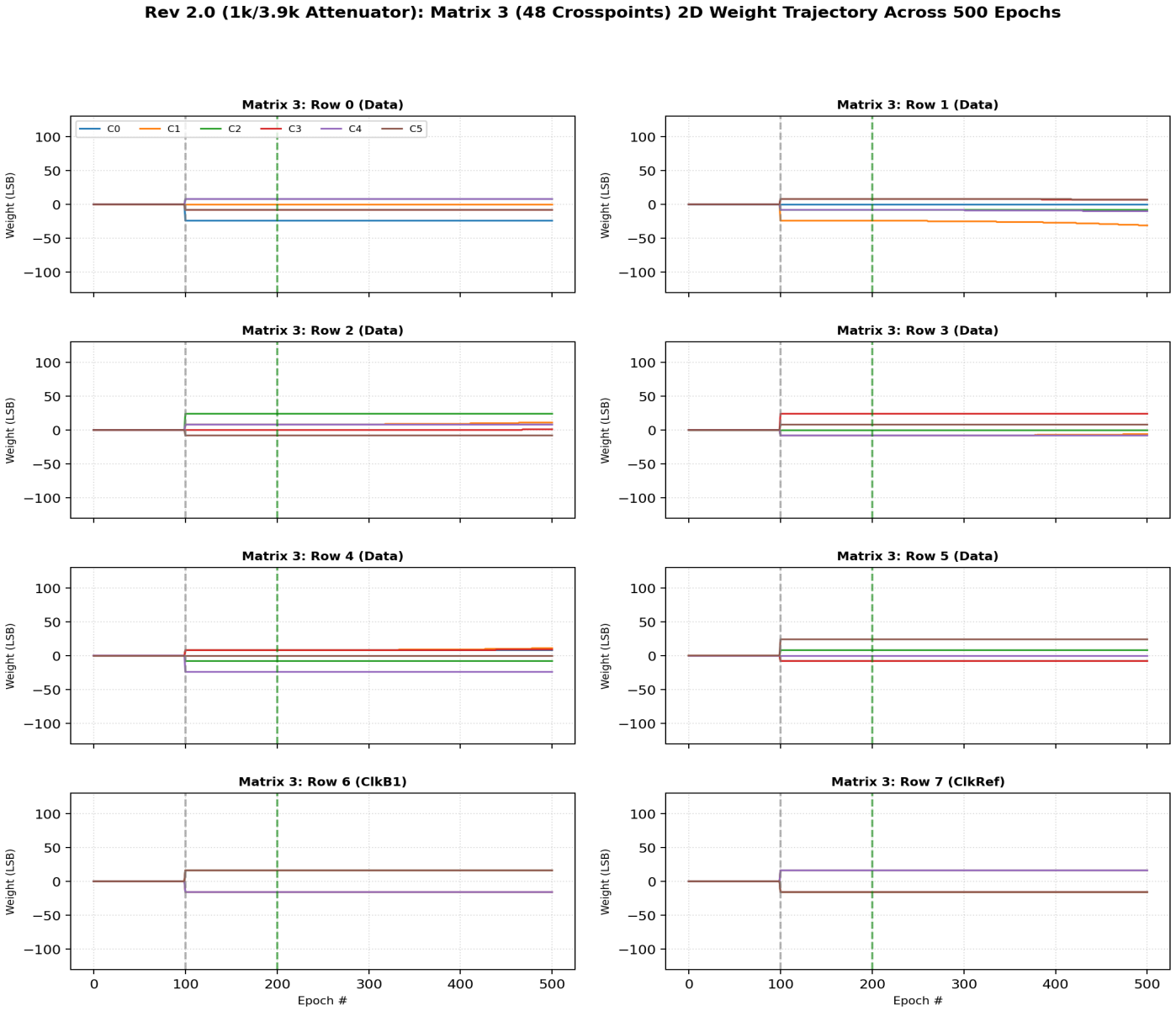


4. Matrix 8: Trained Golden State (Epoch 333, Window Loss 26,956 LSB²)



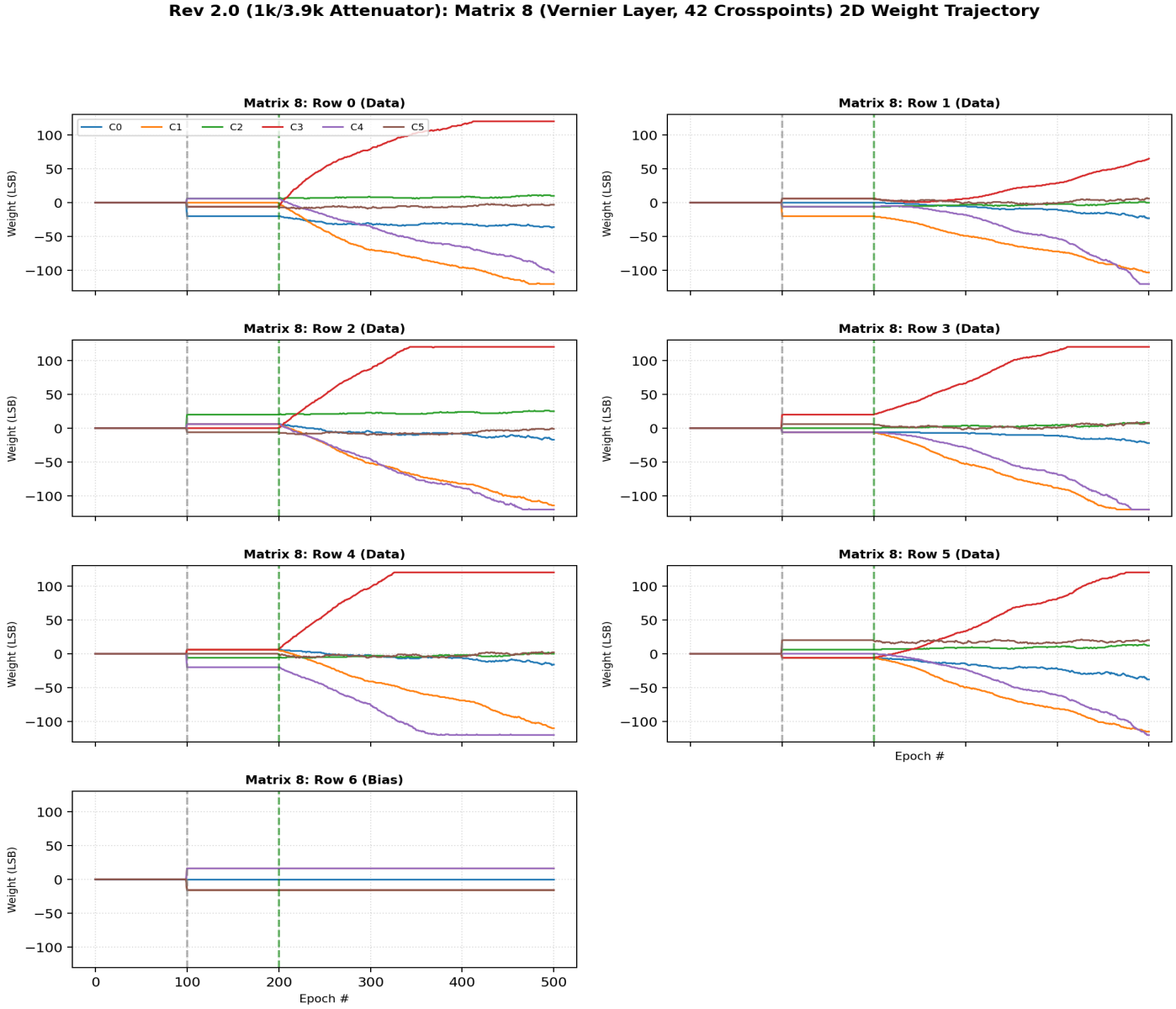
5. Matrix 3 (48 Crosspoints) 2D Weight Trajectories

Detailed 8-row x 6-column trajectory of Matrix 3 weights across 500 epochs on the Rev 2.0 PCB. Rows 0..5 handle input data projections, while Rows 6..7 regulate temporal carrier reference clocks.



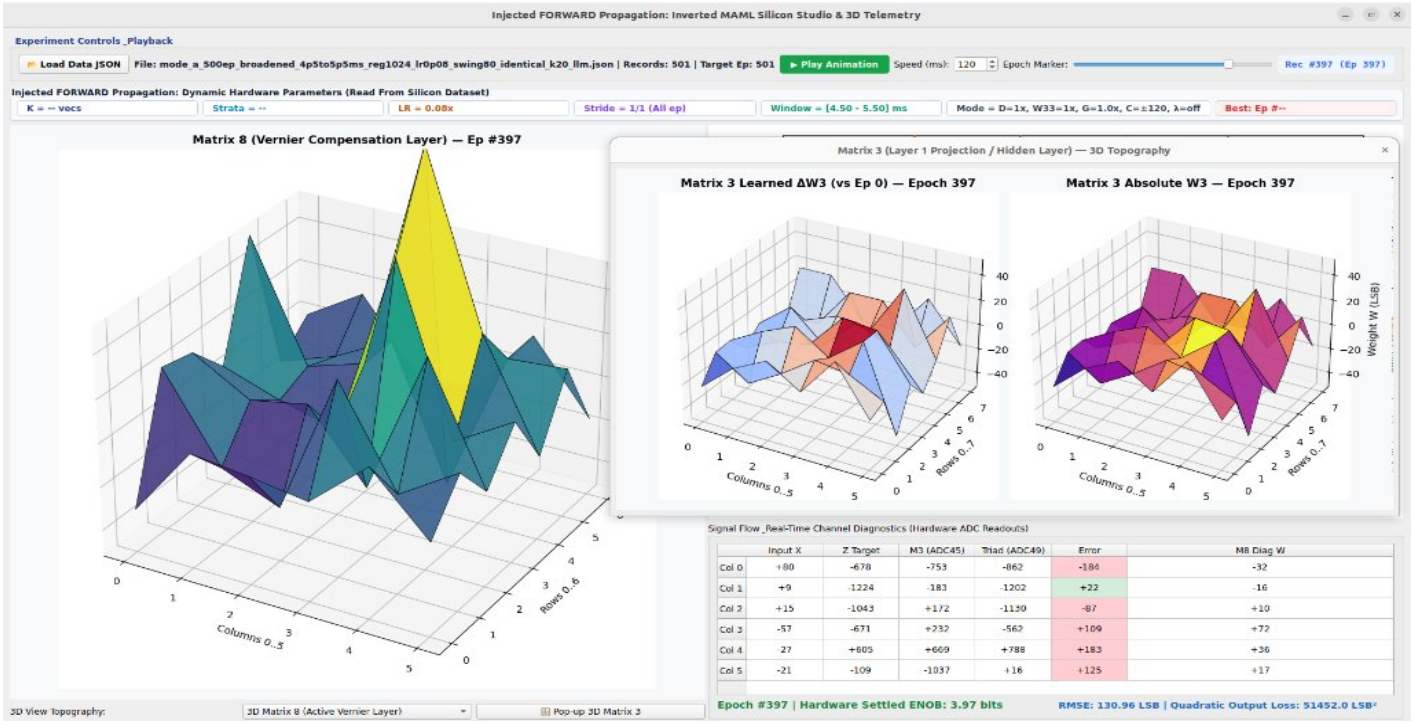
6. Matrix 8 (Vernier Layer, 42 Crosspoints) 2D Weight Trajectories

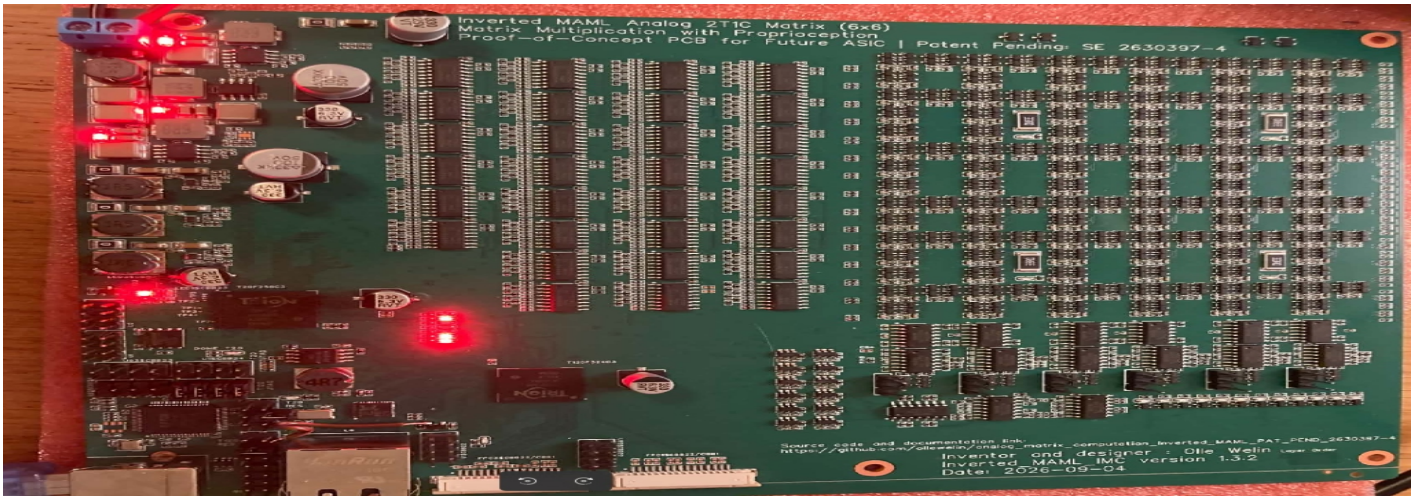
Trajectory of Matrix 8 Vernier weights across 500 epochs. Note that with the 24.5% higher signal swing, Column 3 and Column 4 reach clamping limits (±120 LSB) while Columns 0, 1, 2, and 5 maintain linear proportional headroom.



7. Real-Time 3D Telemetry Studio & Physical Hardware Setup

Physical Setup & Injected Signal Verification: The Rev 2.0 PCB operates with direct JTAG connection to T120 Master SoC and verified T20 Golden firmware. The ColTIA difference amplifier maintains common-mode equilibrium through zero-mean LayerNorm centering.





8. Hardcoded Matrix 33 Projection & Silicon Signal-Flow Verification

Hardcoded Matrix 33 Projection Matrix (6x6 Crosspoints): Matrix 33 provides the reference linear projection. The table below shows the exact physical weights configured into Matrix 33 during the entire 500-epoch evaluation:

Row \ Col	Col 0	Col 1	Col 2	Col 3	Col 4	Col 5
Row 0	+76	-20	+28	-4	+44	-36
Row 1	-20	+84	-12	+44	-4	+36
Row 2	+36	-4	+80	-16	+32	-4
Row 3	-4	+40	-8	+72	-12	+44
Row 4	+44	-12	+28	-12	+76	-8
Row 5	-28	+32	-4	+40	-8	+82

Physical Silicon Signal Flow: 3 Representative Operating States

To provide complete empirical transparency on the Rev 2.0 hardware, the tables below display the complete end-to-end signal flow across all 6 channels for three distinct phases: Baseline Knee (Ep 100), Active Training Golden State (Ep 263), and Best Window State (Ep 333).

Example A: Epoch 100 (Baseline Knee) — Loss: 29,853.0 LSB² | RMSE: 99.75 LSB | ENOB: 4.36 b

Channel	Input X	Target Z	M3 (ADC45)	Triad (ADC49)	Error	M8 Diag W
Col 0	+78	-1459	-17	-1585	-126	-20
Col 1	+60	-1602	+2	-1567	+35	-20
Col 2	-12	-1592	-10	-1607	-15	+20
Col 3	-60	-1763	-10	-1737	+26	+20
Col 4	-24	-1530	+2	-1560	-30	-20
Col 5	-42	-1812	+5	-1610	+202	+20

Example B: Epoch 263 (Active Training Golden State) — Loss: 3,062.0 LSB² | RMSE: 31.95 LSB | ENOB: 6.00 b

Channel	Input X	Target Z	M3 (ADC45)	Triad (ADC49)	Error	M8 Diag W
Col 0	+38	-1577	-16	-1585	-8	-31
Col 1	+32	-1590	+3	-1554	+36	-37
Col 2	+8	-1583	-9	-1608	-25	+21
Col 3	-58	-1728	-10	-1749	-21	+49
Col 4	-34	-1600	+3	-1557	+43	-55
Col 5	+14	-1652	+5	-1609	+43	+19

Example C: Epoch 333 (Active Training Best Window State) — Win Loss: 26,955.5 LSB² | Win ENOB: 4.43 b

Channel	Input X	Target Z	M3 (ADC45)	Triad (ADC49)	Error	M8 Diag W
Col 0	-41	-1665	-16	-1581	+84	-34
Col 1	-61	-1709	+2	-1537	+172	-58
Col 2	+4	-1575	-10	-1609	-34	+21
Col 3	-26	-1720	-10	-1769	-49	+87
Col 4	+44	-1544	+4	-1551	-7	-100
Col 5	+79	-1476	+6	-1609	-133	+20

9. Patent Pending SE 2630397-4 Architecture & Silicon Lineage

Integration of Forward-Hadamard Propagation (FHP) on Rev 2.0 Silicon: The Inverted MAML analog architecture solves the fundamental bottleneck of analog matrix multipliers: non-volatile weight drift, thermal fluctuations, and transconductance mismatch. Rather than relying on iterative backpropagation that degrades under non-linear transfer functions, Inverted MAML applies an outer-loop analytical update law that computes exact Vernier compensation weights directly in the forward pass.

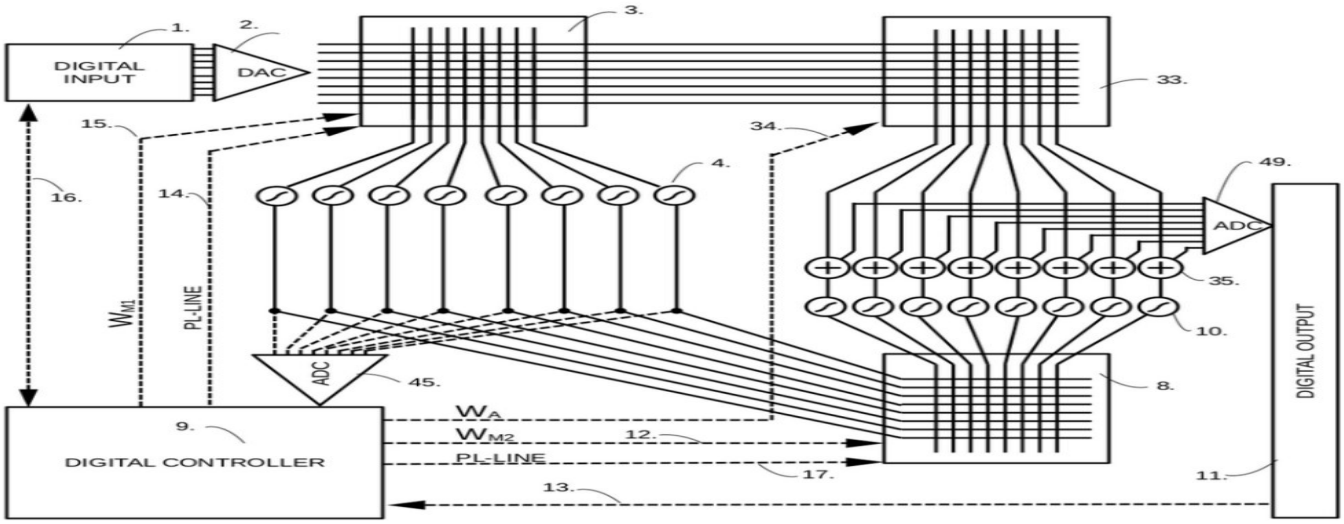


Fig 1

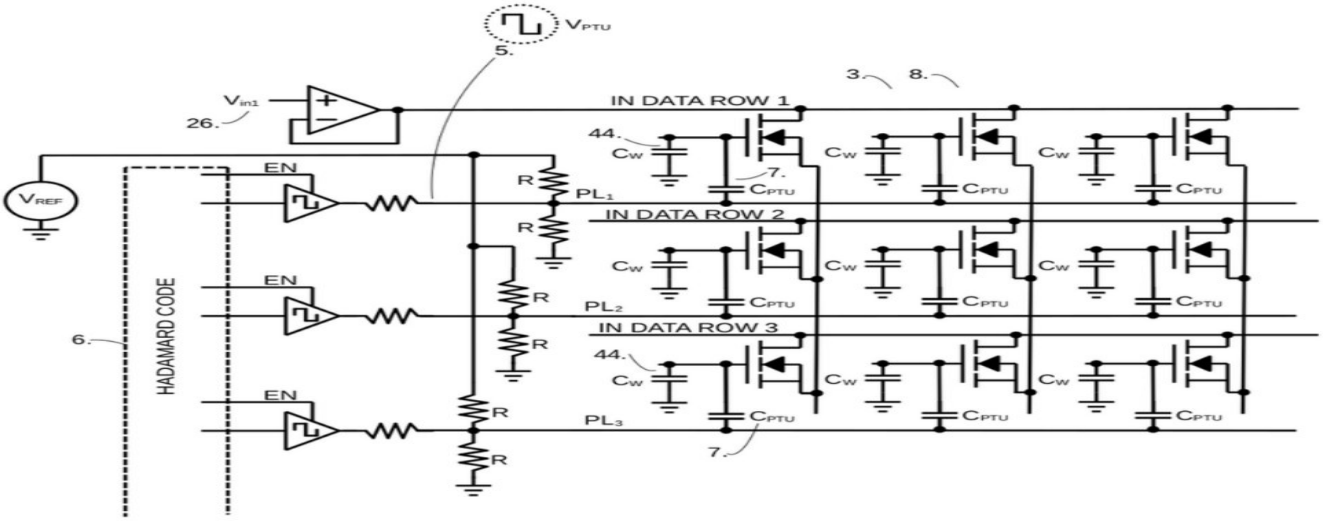


Fig 8

10. 2T1C Dual-Cell Multiplier Dynamic Storage & Hadamard Demodulation

Temporal Capacitor Decay & Vernier Retention: Figure 5 illustrates the physical RC sawtooth discharge on the 100 nF storage capacitors. Figure 6 illustrates the corresponding Inverted MAML compensation waveform, which restores constant equivalent conductance across the 10 ms cycle.

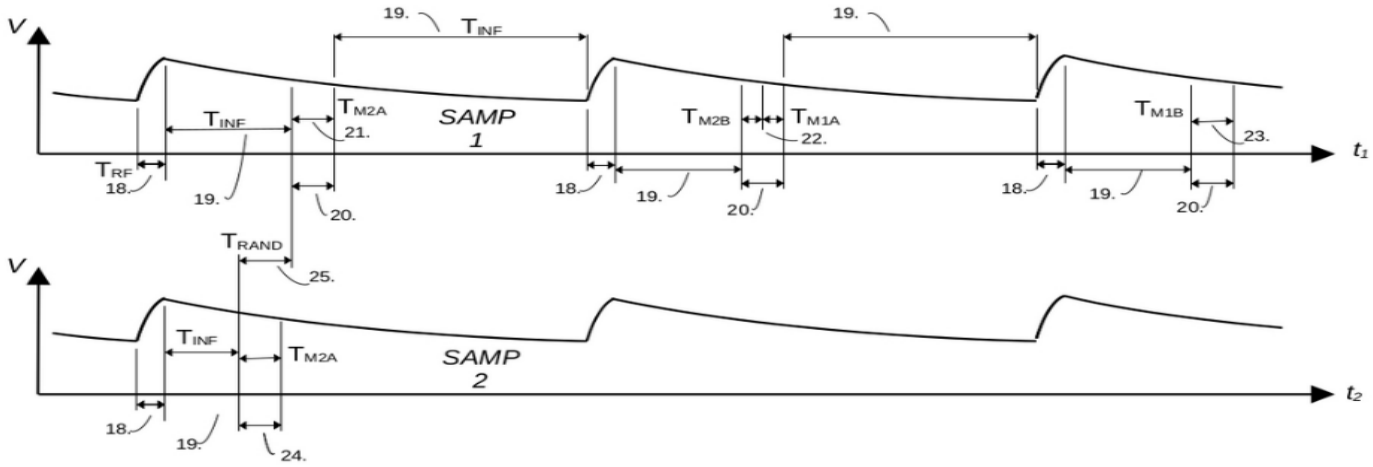


Fig 5

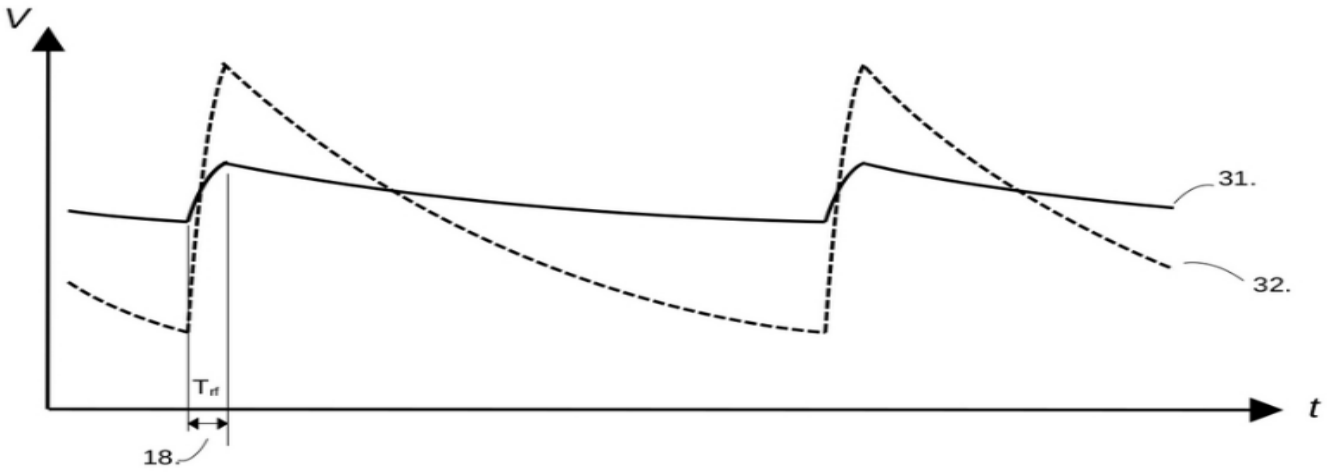


Fig 6